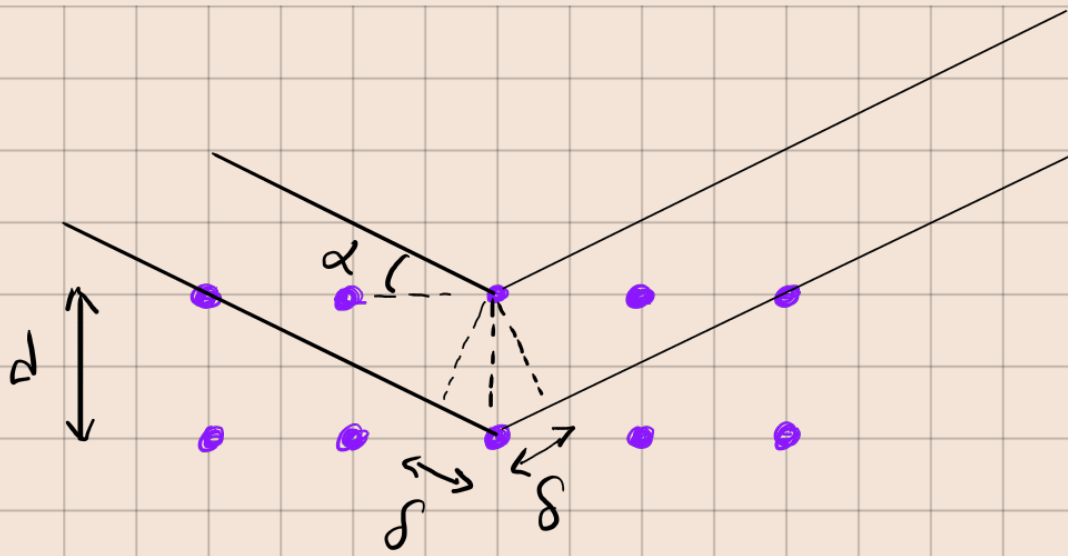
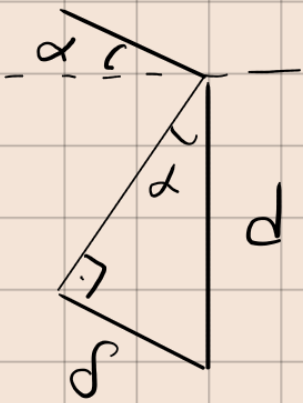


Ex 33



Path difference: $\Delta s = 2\delta$



$$\delta = d \cdot \sin \alpha$$

For constructive interference, $\Delta s = n\lambda$

$$\Rightarrow 2d \cdot \sin \alpha = n\lambda \quad \Rightarrow \quad d = \frac{n\lambda}{2 \sin \alpha}$$

Using the first peak: $n=1$, $\alpha \approx 7.5^\circ$

$$d = \frac{7 \times 10^{-11} \text{ m}}{2 \cdot \sin(7.5^\circ)} = 2.7 \times 10^{-10} \text{ m}$$

Note: using $n=2$, $\alpha \approx 15^\circ$ gives the same result.

Ex 34

Theoretical magnification of a microscope (script 404-10):

↗ optical tube length
↘ viewing distance

$$\Gamma = \frac{s \cdot s_0}{f_1 \cdot f_2}$$

↘ objective focal length
↘ eyepiece focal length

Distance between lenses: $a = s + f_1 + f_2$

a) Maximum a is given as 20 cm.

$$s = a - f_1 - f_2 = 16.8 \text{ cm}$$

$$\Rightarrow \Gamma = \frac{(16.8 \text{ cm}) \cdot (25 \text{ cm})}{(2 \text{ cm}) \cdot (1.2 \text{ cm})} = 175$$

b) Isolate f_1 from the equation

$$\Gamma = \frac{s \cdot s_0}{f_1 \cdot f_2} = \frac{(a - f_1 - f_2) \cdot s_0}{f_1 \cdot f_2}$$

$$\Rightarrow \Gamma f_1 f_2 = (a - f_2) s_0 - f_1 s_0$$

$$\Rightarrow f_1 (\Gamma f_2 + s_0) = (a - f_2) s_0$$

$$\Rightarrow f_1 = \frac{(a - f_2) s_0}{\Gamma f_2 + s_0}$$

Use previous values and $\Gamma = 100$

$$\Rightarrow f_1 = 3.24 \text{ cm}$$

c) We need a magnification of

$$\Gamma = \frac{18.75 \text{ mm}}{2.5 \text{ mm}} = 7.5$$

Then, isolate f_2 (same as previous, just swap f_1 and f_2)

$$f_2 = \frac{(a - f_1) s_0}{\Gamma f_1 + s_0} = 11.25 \text{ cm}$$

Ex 35

script 504-9

Formula for intensity maxima in a diffraction grating experiment:

$$\sin \theta_m = m \frac{\lambda}{d}, \quad m = 0, \pm 1, \pm 2, \dots$$

Spacing: $d = \frac{1 \text{ cm}}{1000} = 10^{-5} \text{ m}$

a) 1st order: $m = 1$ and $\lambda = 650 \text{ nm}$

$$\Rightarrow \theta_1 = \arcsin \left(\frac{\lambda}{d} \right) = 3.73^\circ$$

b) $m = \frac{d}{\lambda} \cdot \sin \theta_m$

m is maximum when $\sin \theta_m = 1$

$$\Rightarrow m_{\max} = \frac{d}{\lambda} = 15.38$$

m must be an integer, so we take

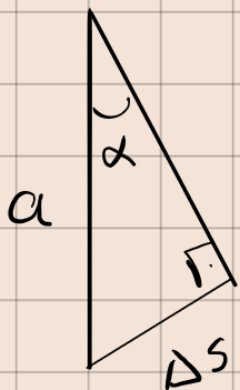
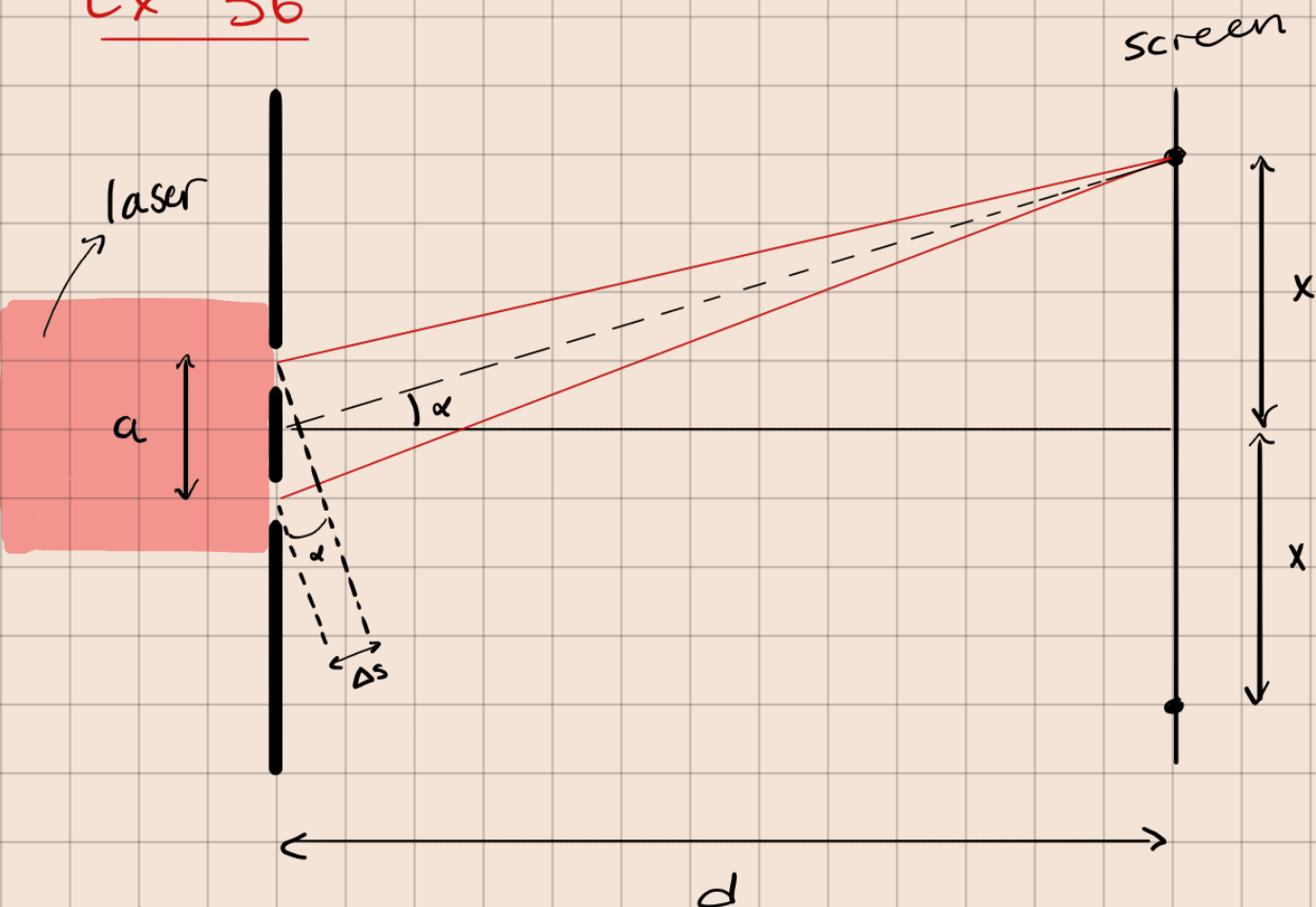
$$m = 15$$

c) Use $m=1$ and rearrange equation.

$$\lambda = d \cdot \sin \theta$$

- for $\theta = 2.46^\circ$, $\lambda = 429.2 \text{ nm}$
- for $\theta = 3.15^\circ$, $\lambda = 549.5 \text{ nm}$

Ex 36



$$\sin \alpha = \frac{\Delta s}{a}$$

$$\tan \alpha = \frac{x}{d} \approx \sin \alpha$$

Intensity maxima: $\Delta s = n \cdot \lambda$

We know total distance between 4th order maxima.

$$2x = 26 \text{ mm} \Rightarrow x = 13 \text{ mm}$$

Use $n = 4$:

$$\Rightarrow \Delta s = 4\lambda$$

$$\Rightarrow a \cdot \sin \alpha = 4\lambda$$

$$\Rightarrow a \cdot \frac{x}{d} = 4\lambda$$

$$\Rightarrow a = \frac{4\lambda d}{x} = \frac{4 \cdot (633 \text{ nm}) \cdot (1700 \text{ mm})}{13 \text{ mm}}$$

$$= 331.1 \text{ } \mu\text{m}$$